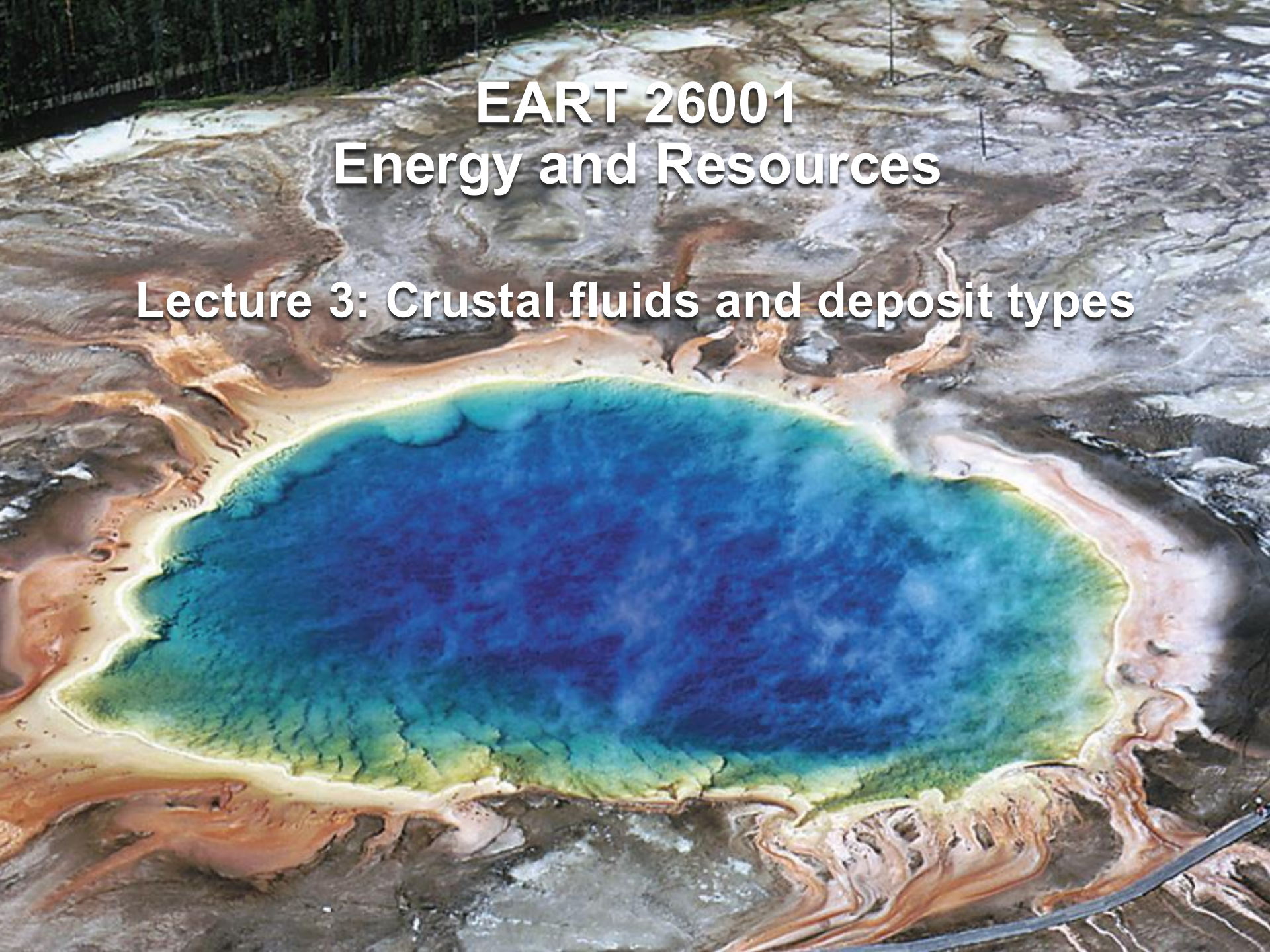


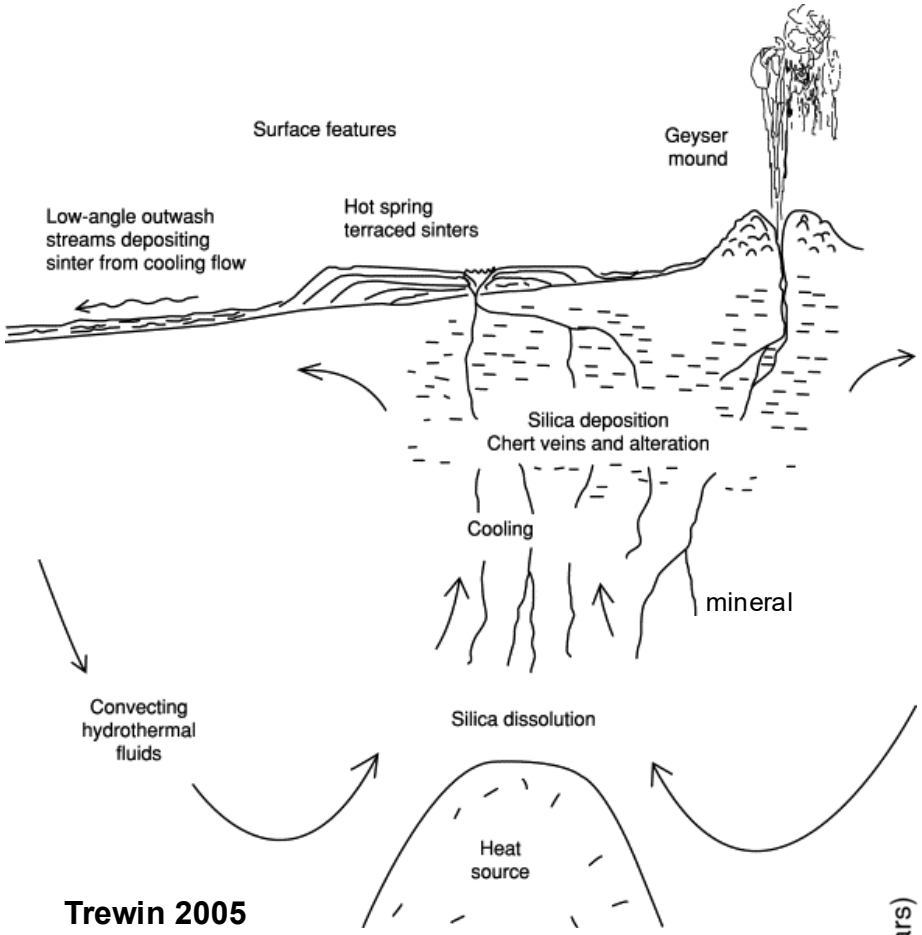
EART 26001
Energy and Resources

Lecture 3: Crustal fluids and deposit types

Hydrothermal Deposits and Processes

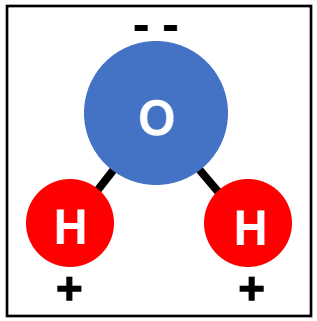


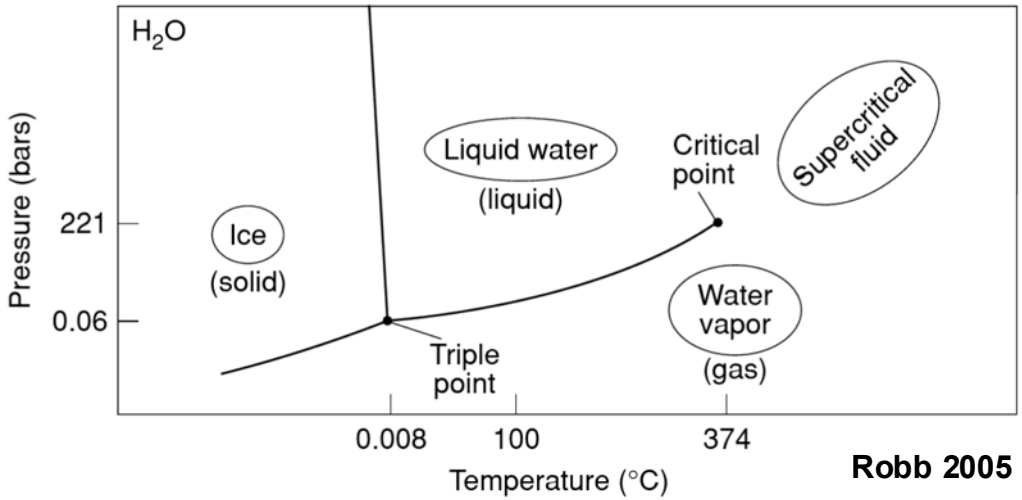
Hydrothermal Deposits and Processes
Hydrothermalism



Trewin 2005

- Involves hot aqueous fluids: thermal energy contained and transmitted by water
- External heat source and circulation in crust (pathways!) -> element concentration
- Key properties of water:
  - Dissolve ions due to dipole moment
  - Presence of Cl^-, HS^-, CO_3^{2-}, SO_4^{2-} as ligands for metal cations
  - High surface tension to wet mineral surfaces
  - High heat conductivity

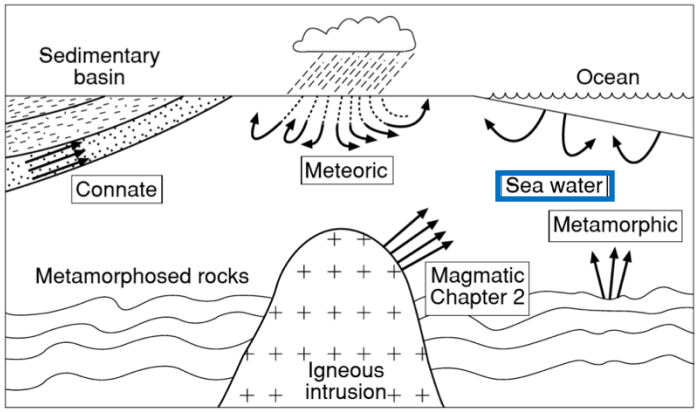




Robb 2005

Hydrothermal Deposits and Processes
Aqueous fluids – sea water





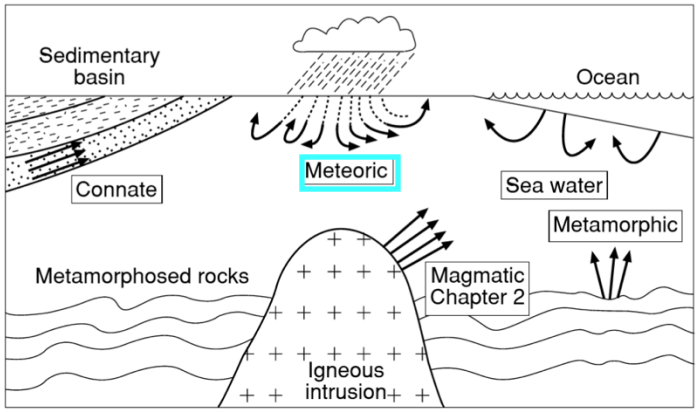
- **Fluid composition:**
  - **Mild salinity**
  - **Na^+, K^+, Mg^{2+}, Cl^-, SO_4^{2-}**
  - **Oxygenated**

	Seawater[b] Total Dissolved Solids (%) (Salinity = 35,000 ppm or 3.5%)
Bicarbonate $(HCO_3)^-$ and carbonate $(CO_3)^{2-}$	0.4
Calcium (Ca^{2+})	1.2
Amorphous silica (H_4SiO_4)	<0.01
Sulfate $(SO_4)^{2-}$	7.7
Chlorine (Cl^-)	55.0
Sodium (Na^+)	30.6
Magnesium (Mg^{2+})	3.7
Potassium (K^+)	1.1
Iron $(Fe^{2+}$ and $Fe^{3+})$	<0.01
Aluminum $Al(OH)_4^-$	<0.01
Nitrate $(NO_3)^-$	<0.01
Total	99.7

[a]Livingston, 1963.
[b]Mason, 1966.

Hydrothermal Deposits and Processes
Aqueous fluids – meteoric water





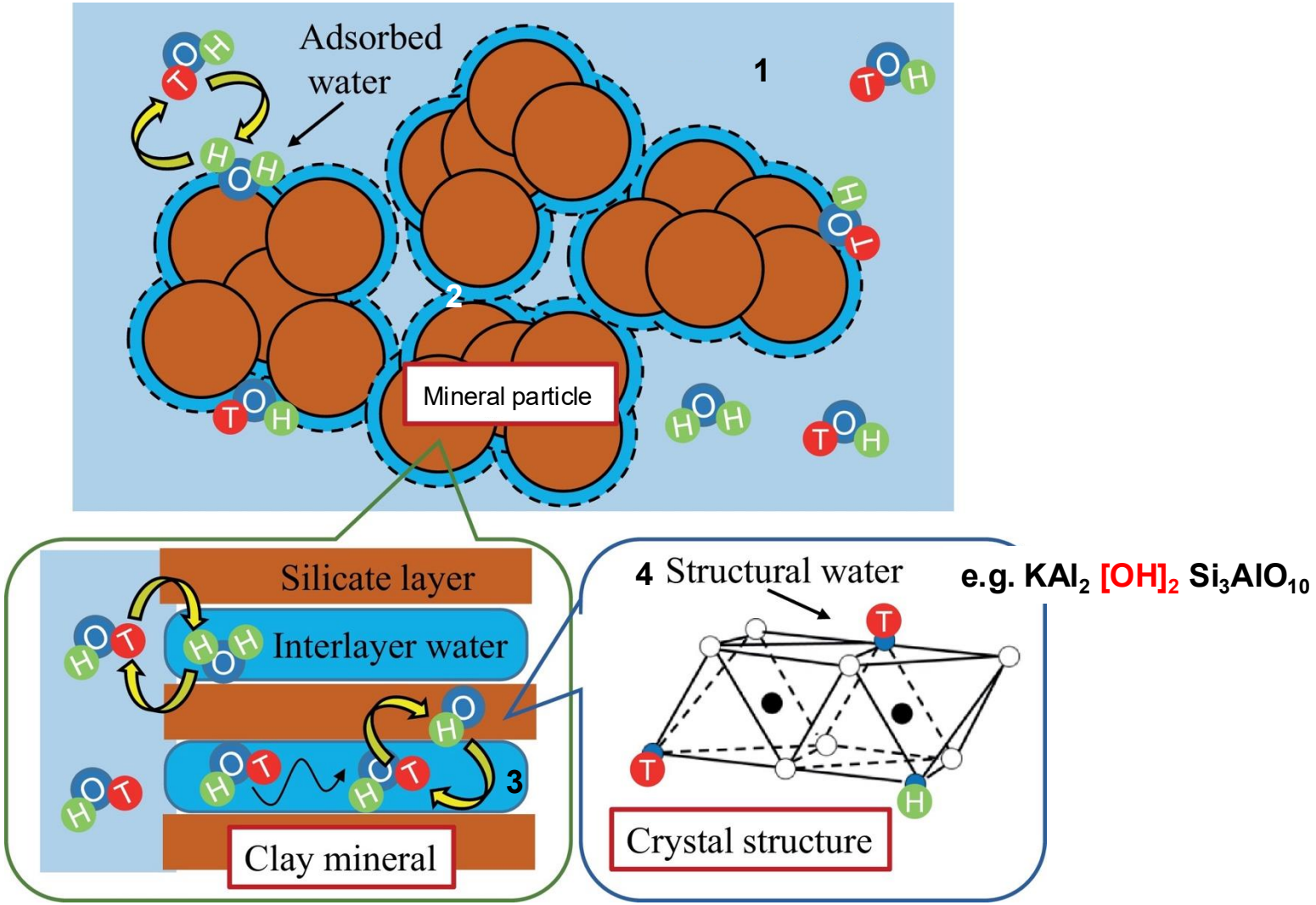
- **Fluid composition:**
  - **Low salinity**
  - **Ca^{2+}, Mg^{2+}, Si^{2+}, HCO_3^-, SO_4^{2-}**
  - **Oxygenated**

	Mean River Water[a] Total Dissolved Solids (%) (Salinity = 121 ppm or 0.0121%)
Bicarbonate $(HCO_3)^-$ and carbonate $(CO_3)^{2-}$	48.6
Calcium (Ca^{2+})	12.4
Amorphous silica (H_4SiO_4)	10.8
Sulfate $(SO_4)^{2-}$	9.3
Chlorine (Cl^-)	6.5
Sodium (Na^+)	5.2
Magnesium (Mg^{2+})	3.4
Potassium (K^+)	1.9
Iron $(Fe^{2+}$ and $Fe^{3+})$	0.6
Aluminum $Al(OH)_4^-$	0.2
Nitrate $(NO_3)^-$	0.8
Total	99.7

[a]Livingston, 1963.
[b]Mason, 1966.

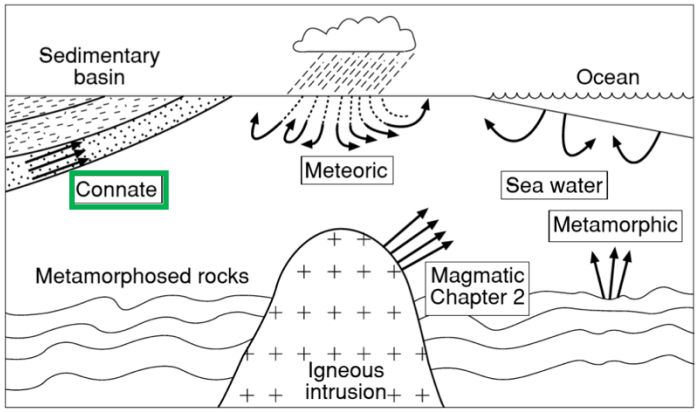
Hydrothermal Deposits and Processes
Aqueous fluids associated with minerals

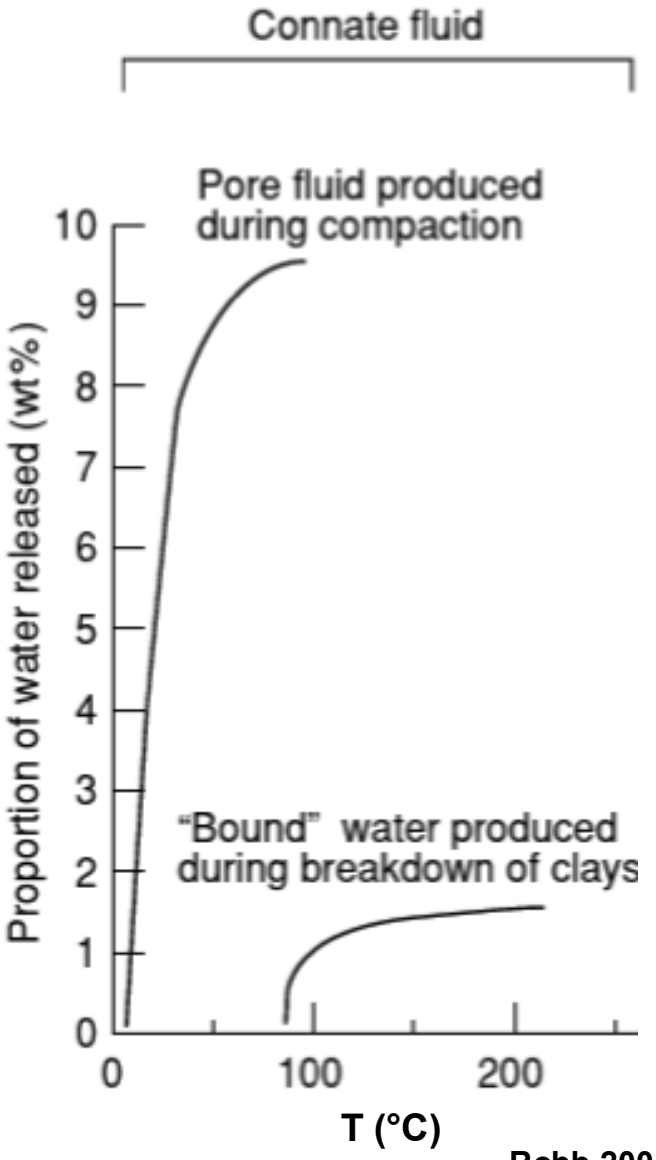
Types of water associated with minerals



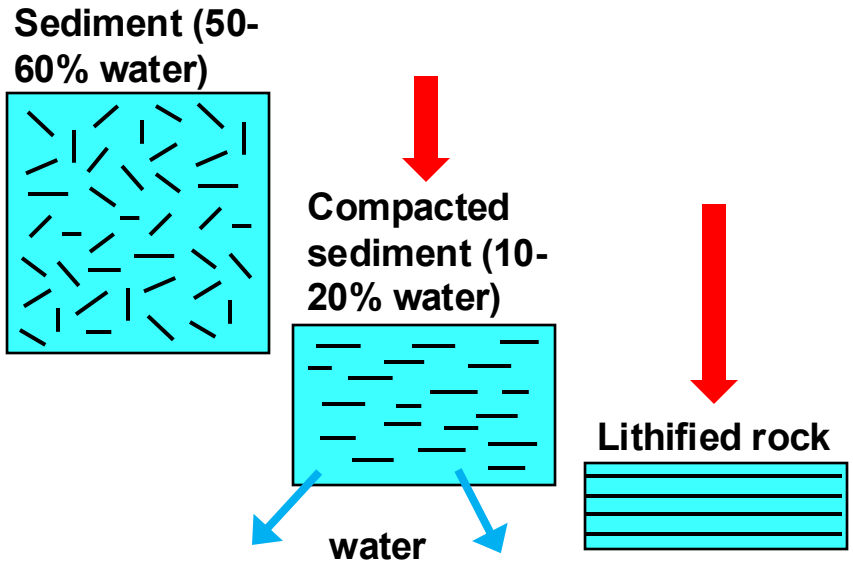
Hyagu 2018

Hydrothermal Deposits and Processes
Aqueous fluids – connate water



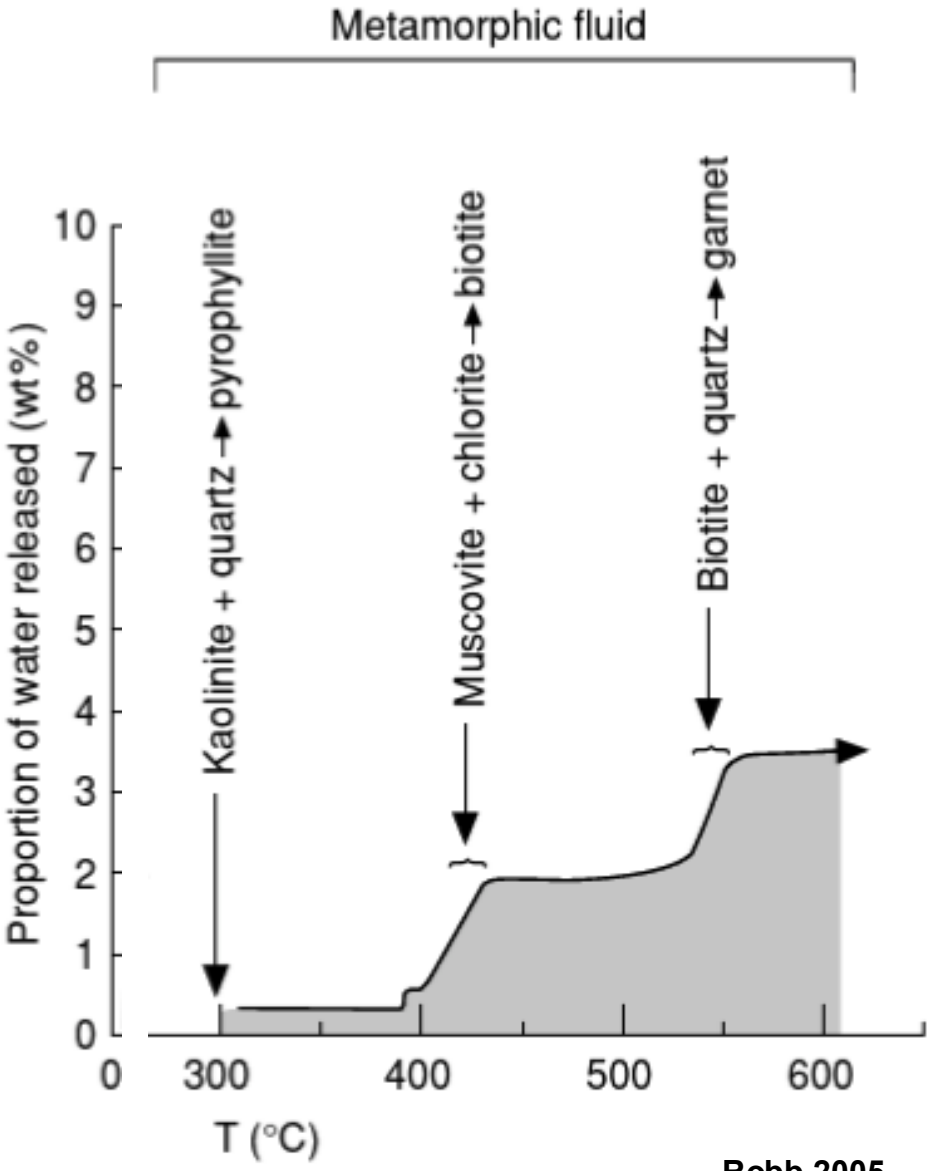


Robb 2005

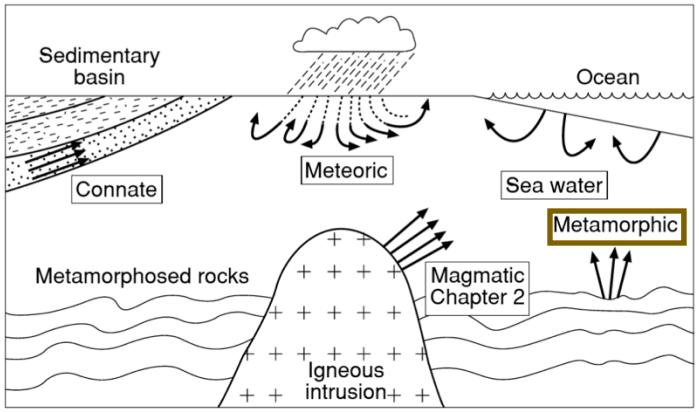


- Clay compaction, dissolution of evaporites
- Fluid composition depending depositional environment: seawater or meteoric water

Hydrothermal Deposits and Processes
Aqueous fluids – metamorphic water

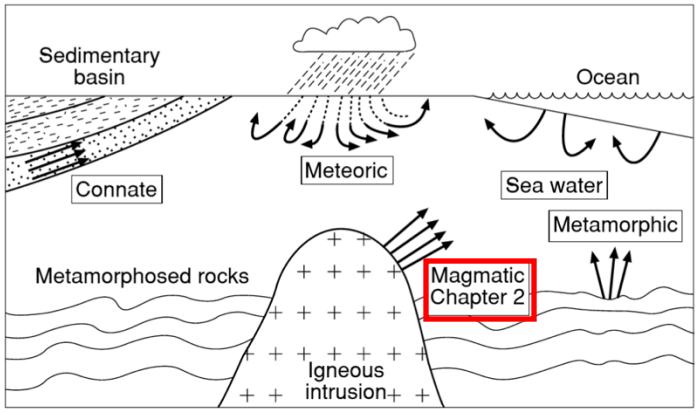


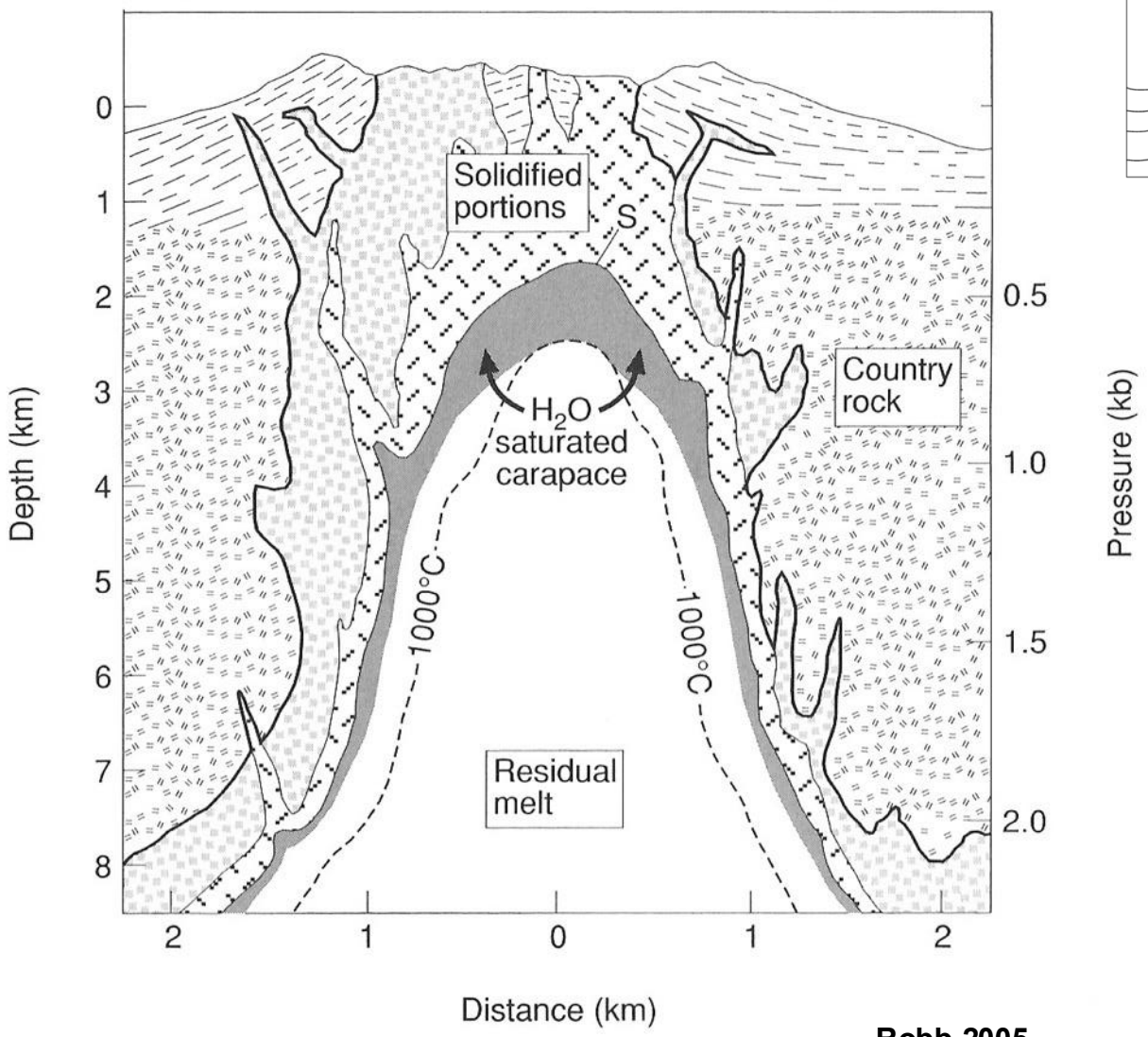
Robb 2005



- Release of structural water as mineral phases adjust to higher pT conditions
- Fluid composition:
  - Low salinity
  - Low sulfur
  - CO_2, CH_4

Hydrothermal Deposits and Processes
Magmatic-hydrothermal fluids

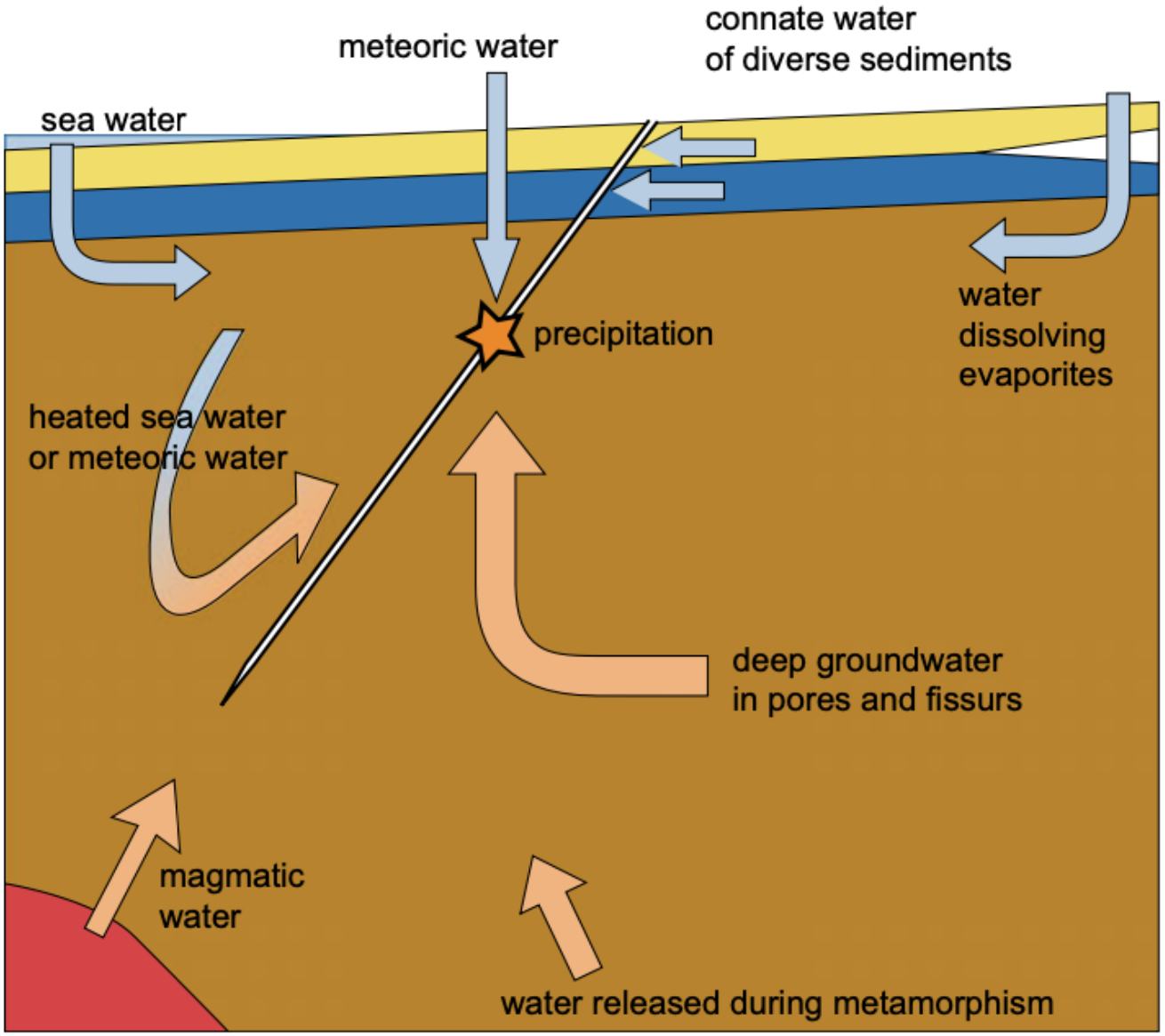




Robb 2005

- **Exsolution from melt**
- **Fluid composition:**
  - **High salinity**
  - **Cl^-, HS^-, CO_3^{2-}, SO_4^{2-}**

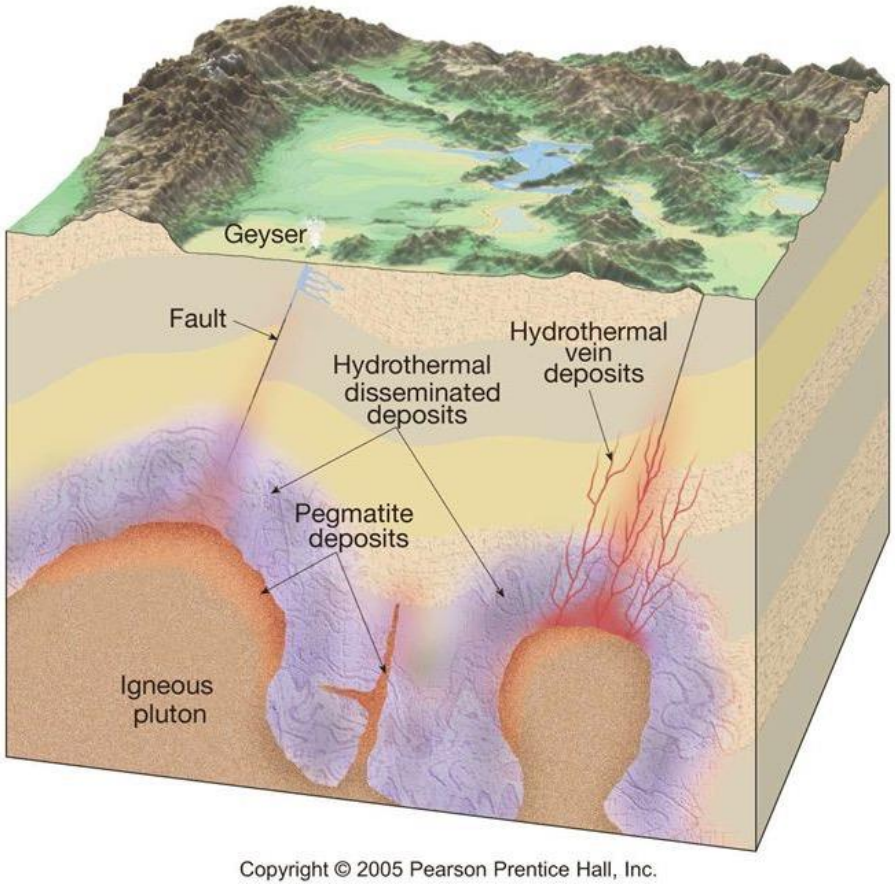
Hydrothermal Deposits and Processes
Hydrothermal system and terminology



Neukirchen 2020

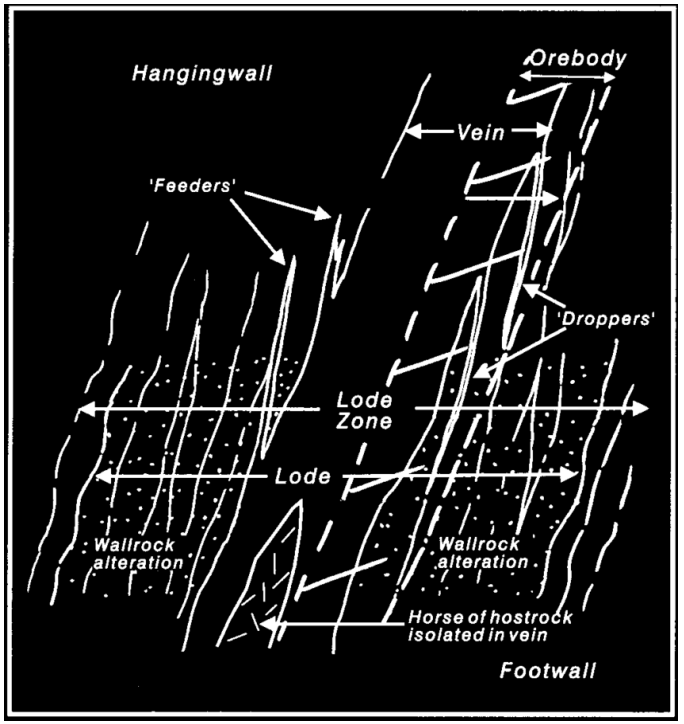
Hydrothermal Deposits and Processes
Hydrothermal system and terminology

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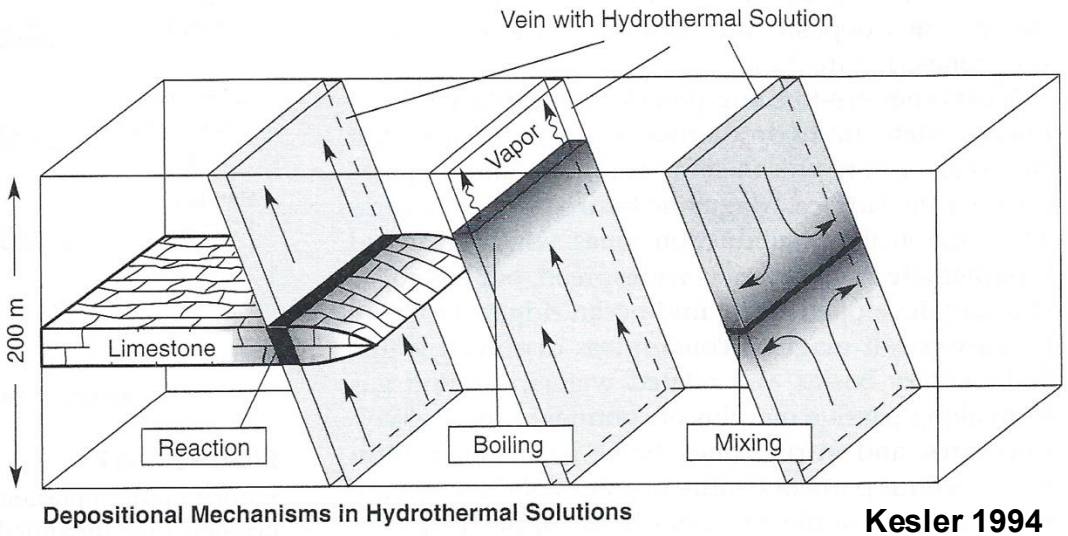


Dominy 1998

- Disseminated
- Vein
- Stockwork

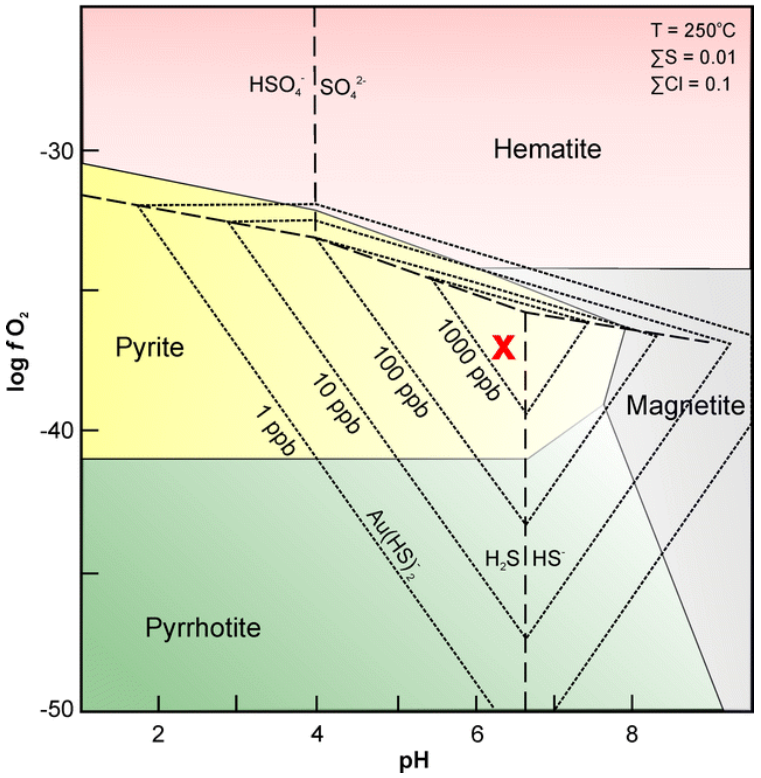
- Gangue
- Shoot/lode

Hydrothermal Deposits and Processes
Precipitation mechanisms



Depositional Mechanisms in Hydrothermal Solutions

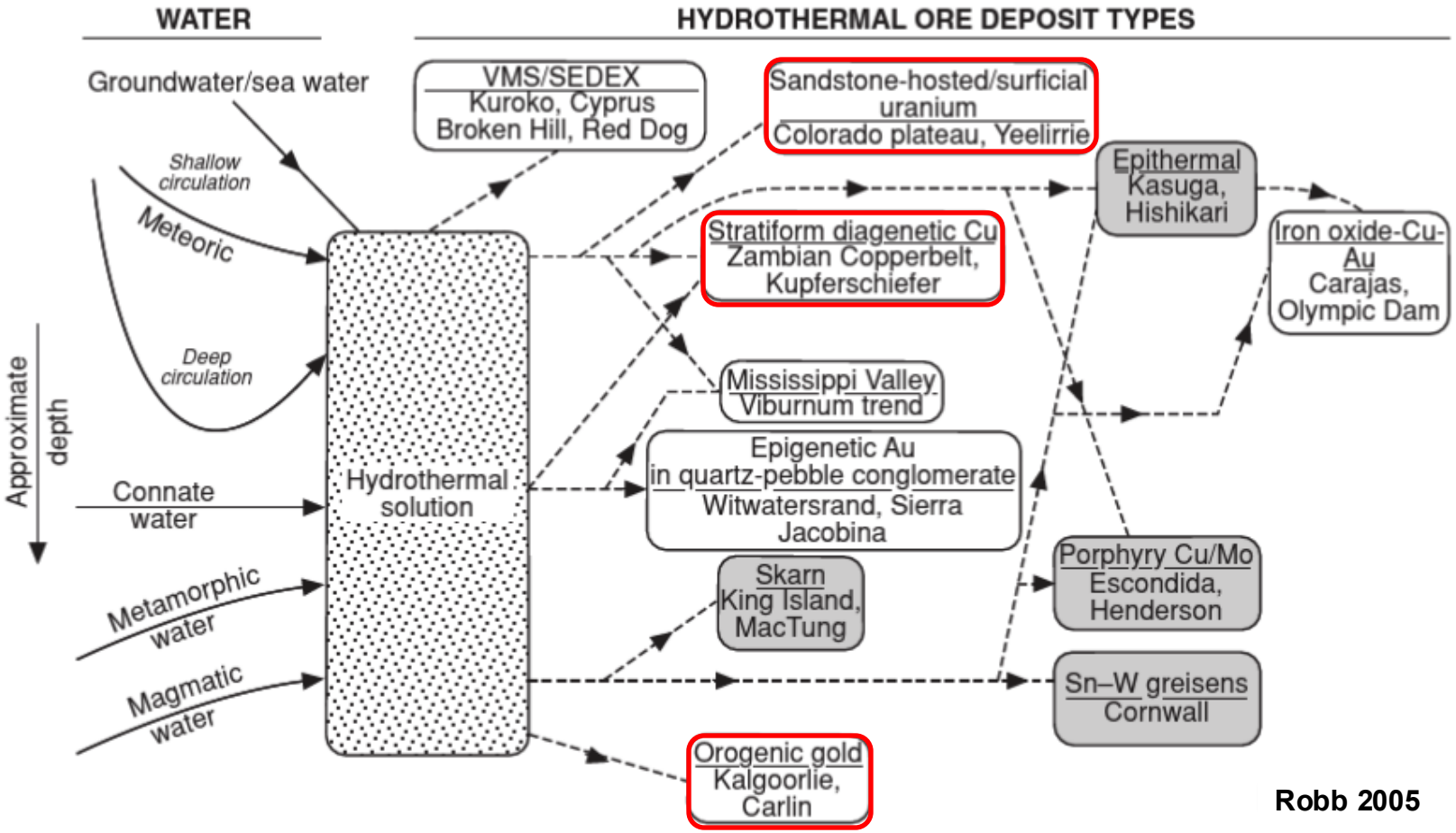
Kesler 1994



Frimmel 2014

- Requires changes to metal solubility, controlled by:
  - Temperature and pressure variation
  - Boiling
  - Fluid mixing
  - pH
  - Redox reactions
- Adsorption on mineral surfaces
- Biological influence
- Ore formation may cement pore space or replace pre-existing minerals

Hydrothermal Deposits and Processes
Deposit overview



Robb 2005

Sedimentary & Surface Deposits and Processes



Sedimentary & Surface Deposits and Processes
Hydrocarbon fluids



Heavy viscous crude

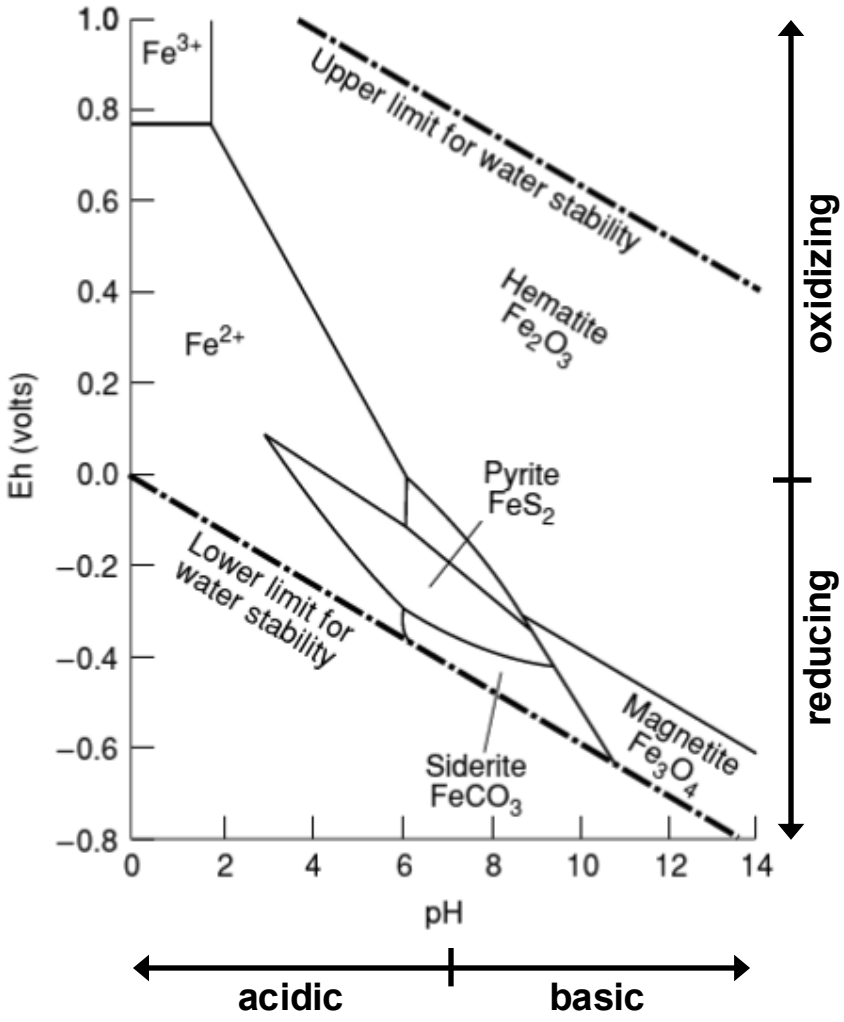
Light sweet crude

condensate

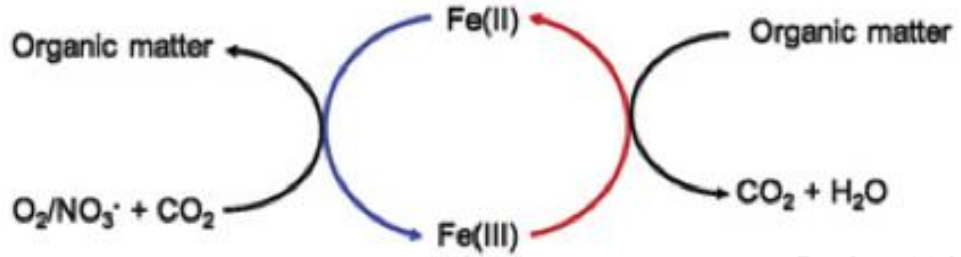


- **Fluid composition:**
  - Aqueous fluid with organic components
  - Various hydrocarbon molecules (C-H)
  - Reduced

Sedimentary & Surface Deposits and Processes
Redox reactions



Robb 2005



Roden 2012

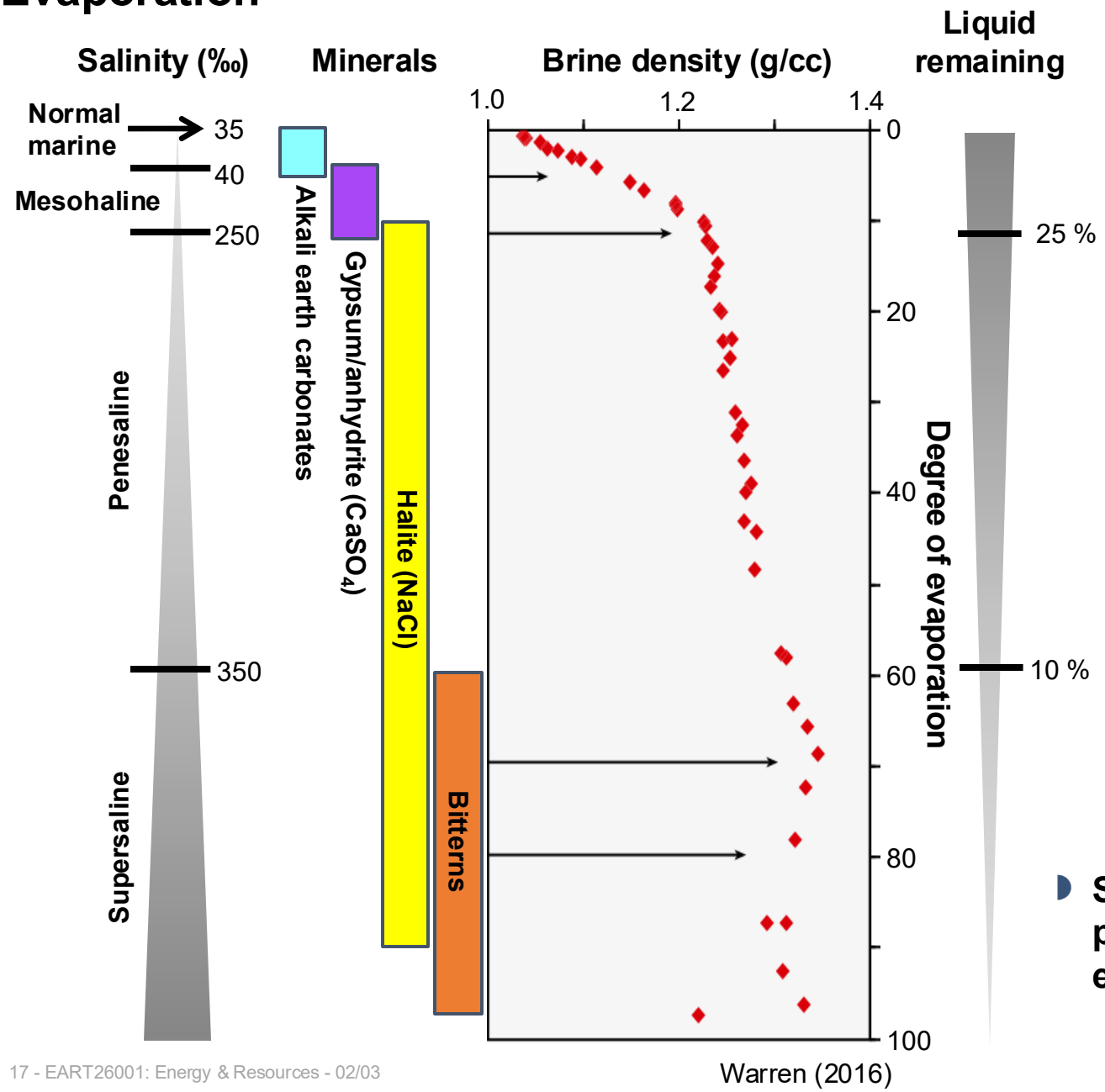




- Allows dissolution, transport, reprecipitation
- In sea water, meteoric water (lakes) and soils
- Often microbially driven

Sedimentary & Surface Deposits and Processes

Evaporation





Neukirchen 2020

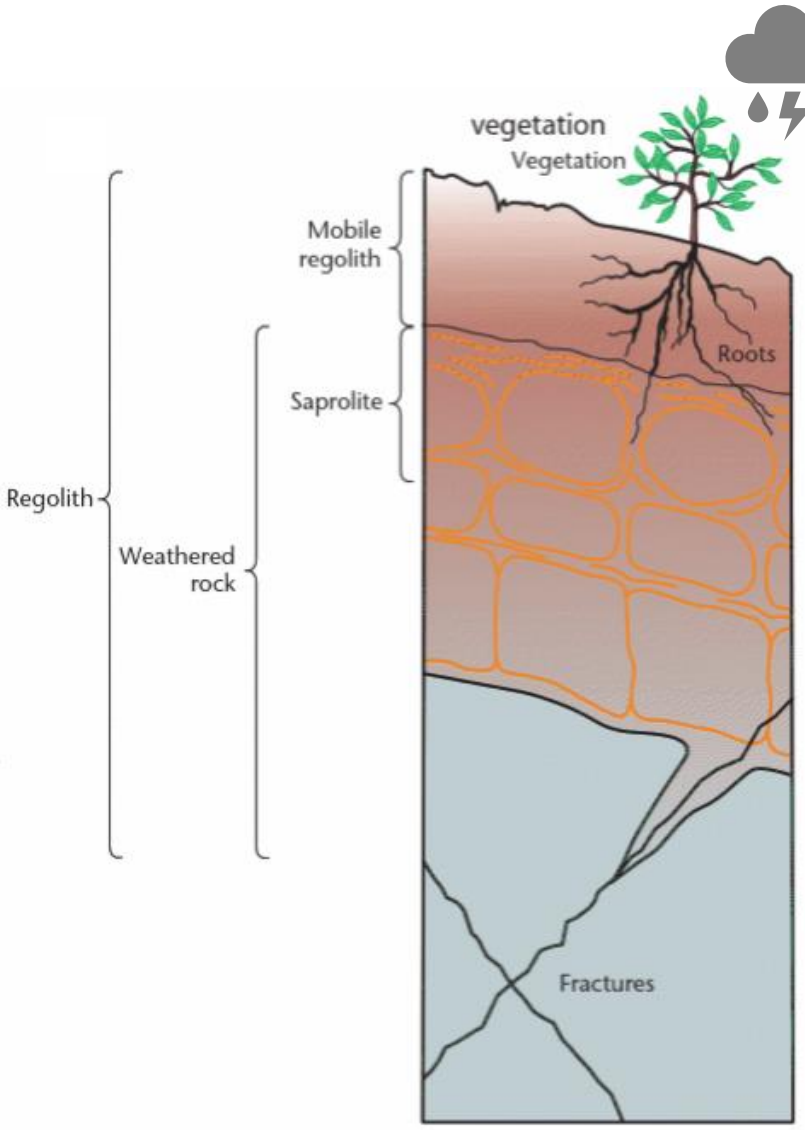


© Reuters/Ivan Alvarado

- Salts dissolved in water precipitate due to (solar) evaporation

Sedimentary & Surface Deposits and Processes
Weathering

- Breakdown of primary rocks/minerals
- Transport of soluble ions
- Formation of new minerals
- Accumulation of less soluble residuals (e.g. Si, Al)
- Processes:
  - Mineral dissolution
  - Mineral hydrolysis
    e.g. $KAlSi_3O_8 + H_2O \rightarrow Al_2Si_2O_5(OH)_4$
  - Oxidation



Cementation at groundwater table

Leaching of less soluble minerals

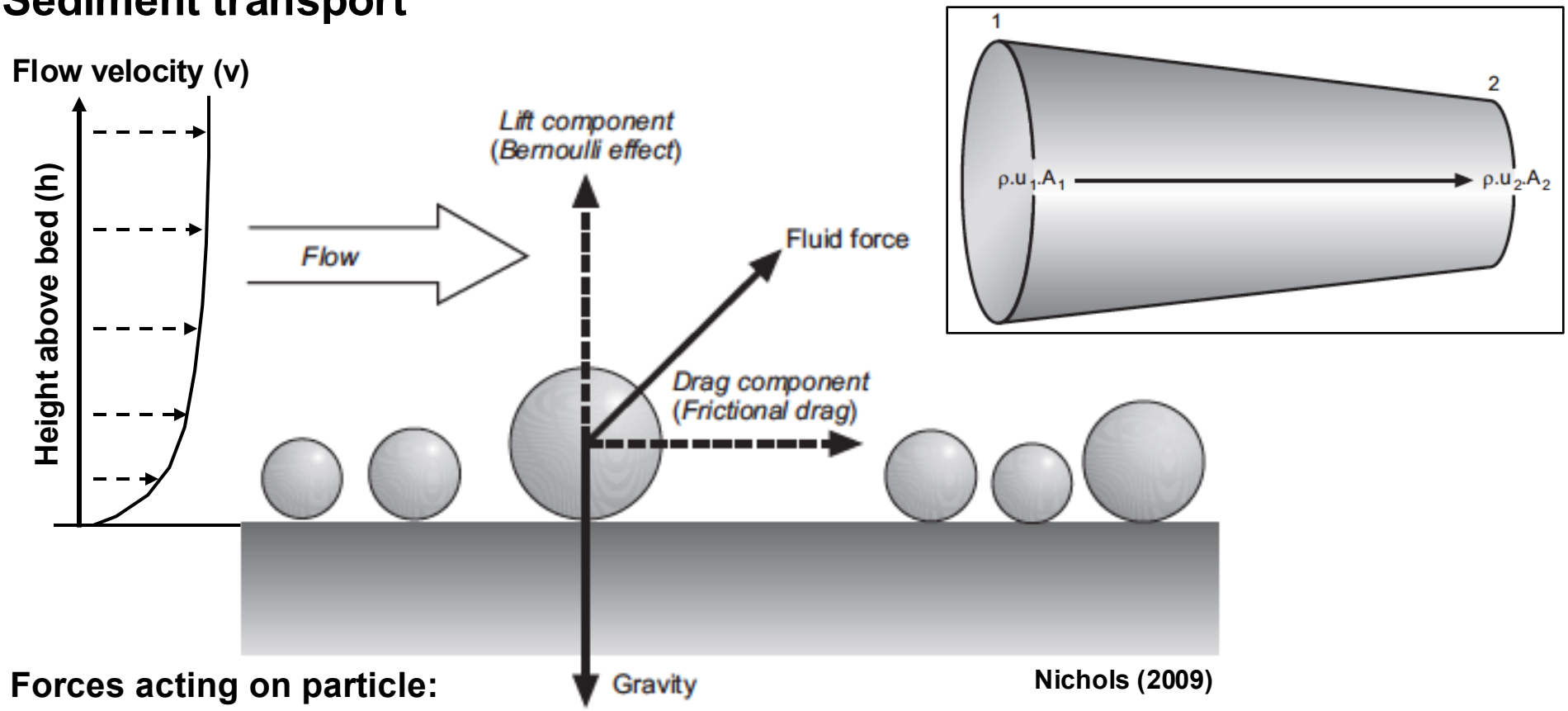
Few primary minerals

Abundant primary minerals e.g. feldspar -> kaolinite, smectite

Weathering of silicates
Carbonate dissolution

Allen 2013

Sedimentary & Surface Deposits and Processes
Sediment transport



Nichols (2009)

Forces acting on particle:

- ✓ Particle mass (gravity)
- ✓ Frictional forces between grains

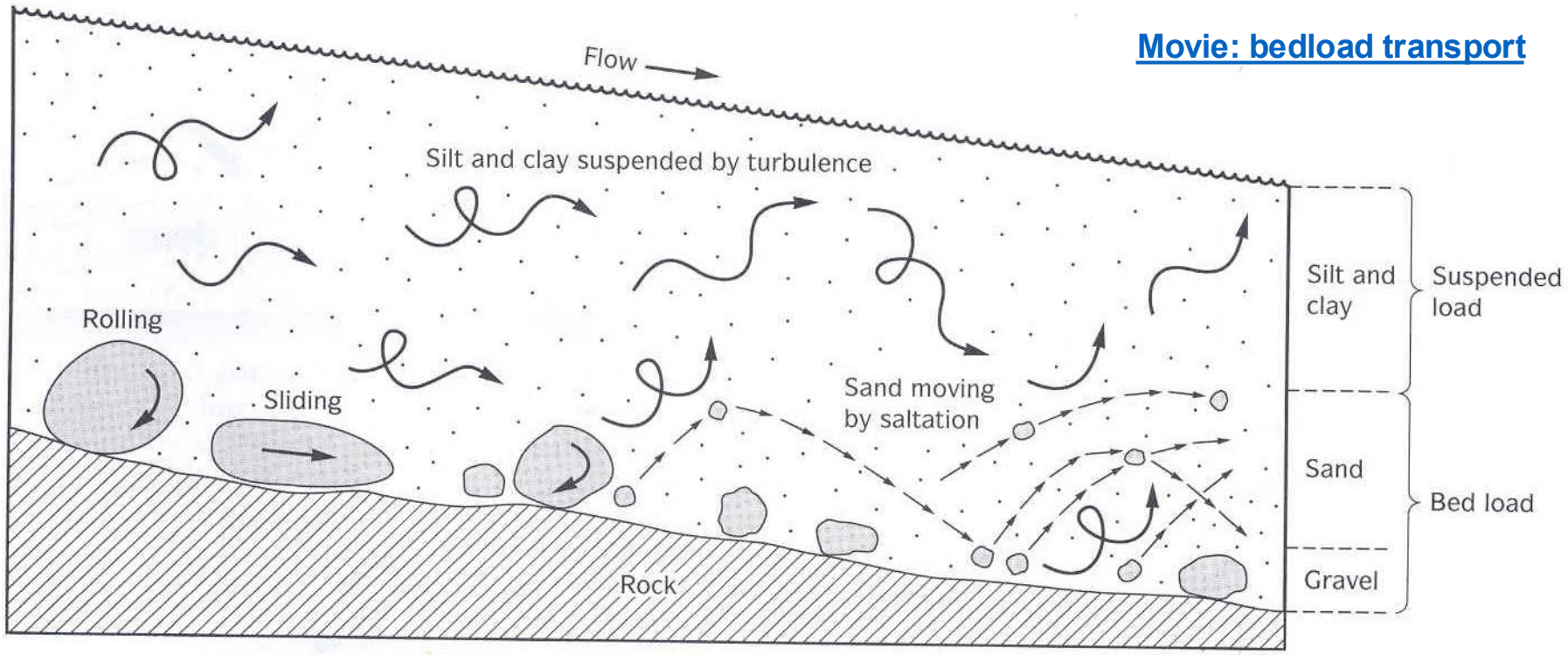
} **Resist movement**

- ✓ Hydraulic lift force
- ✓ Frictional drag

} **Promote uplift and entrainment along bed**

- **Critical flow velocity: Minimum flow velocity to entrain a particle**
- **Rule : Drag and lift forces required will increase with particle mass (~grain size)**

Sedimentary & Surface Deposits and Processes
Sediment transport

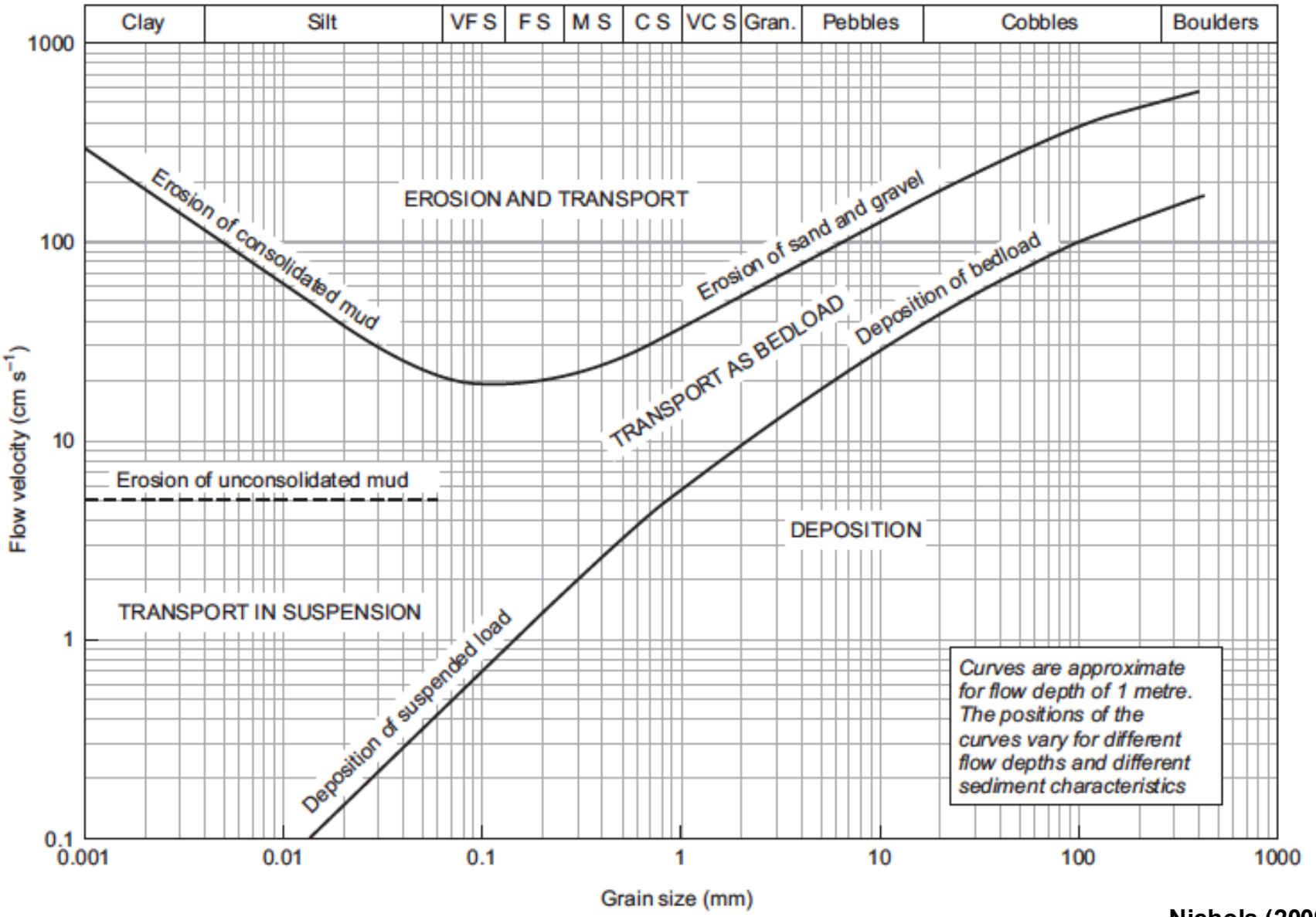


Movie: bedload transport

Nichols (2009)

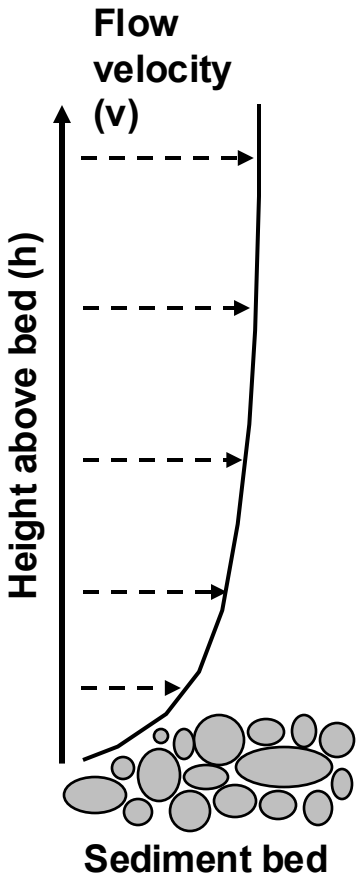
- Bedload transport is main mechanism for clastic sediment transport
- Higher flow velocity needed to move larger grain sizes in suspension

Sedimentary & Surface Deposits and Processes
Sediment transport

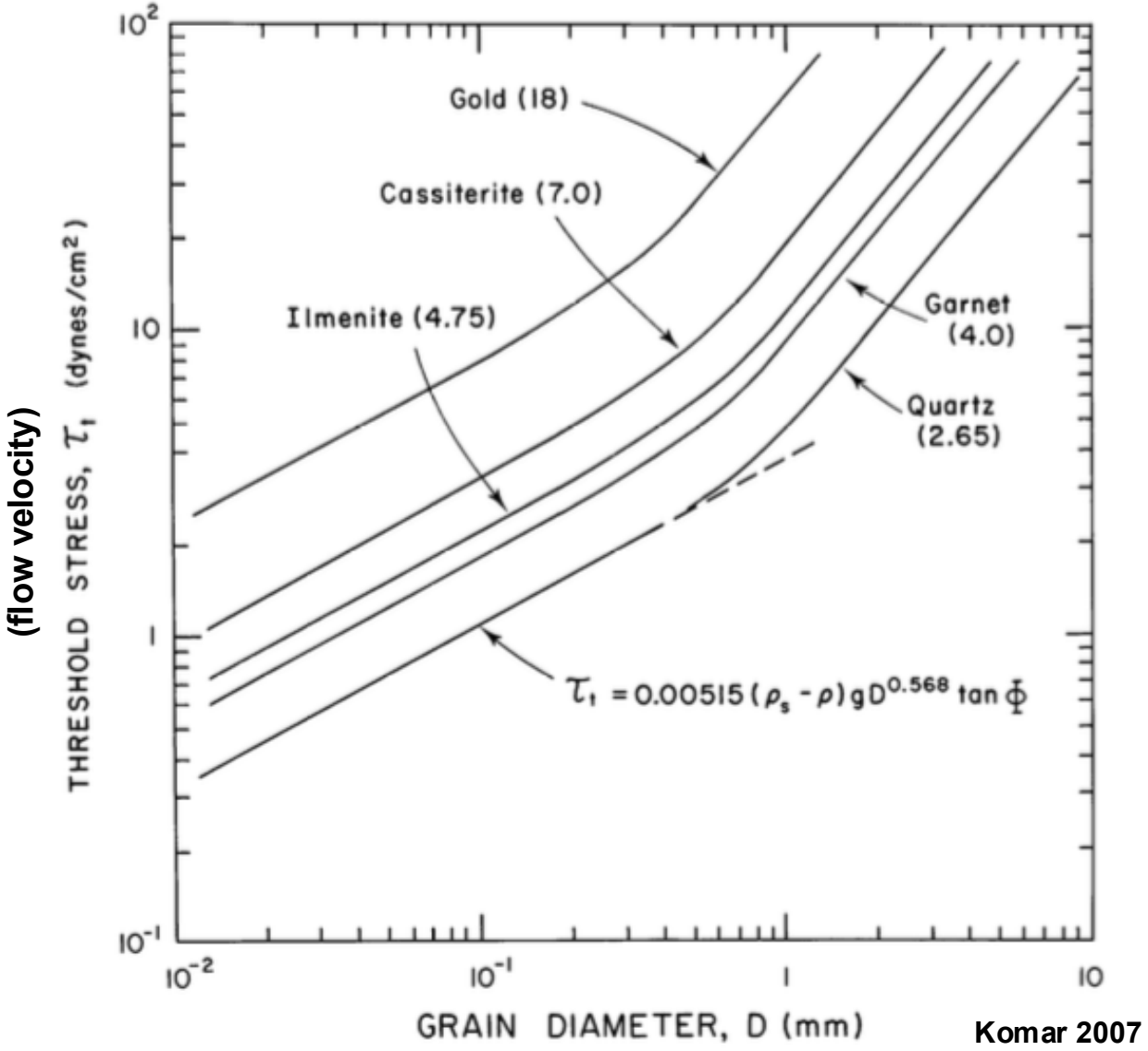


Nichols (2009)

Sedimentary & Surface Deposits and Processes
Sediment transport



Sediment entrainment



Komar 2007

Sedimentary & Surface Deposits and Processes
Sediment transport

Sediment settling (Stoke's Law)

$$V = \frac{(\rho_1 - \rho) \times g \times d^2}{18 \times \mu}$$

V = Settling velocity
ρ_1-ρ = Density difference between particle & fluid
g = Acceleration due to gravity
μ = Viscosity
d = Diameter of particle

- settling velocity **increases** with increasing diameter of particle => 'Large grains fall faster'
- settling velocity **increases** with denser particles:

Mineral	Density (g/cm^3)
Quartz	2.65
Feldspar	2.55
Micas	2.83
Heavy minerals	>3.00
Gold	19



© Prospecting and Mining Journal

Sedimentary & Surface Deposits and Processes
Deposit overview

- Localized element enrichment by processes of sediment transport/deposition, chemical precipitation and weathering
- Supergene: secondary enrichment of existing deposit under near-surface conditions
- Sediment transport and deposition
  - placer deposits of heavy minerals (Au, U, Ti, Zr….)
  - aggregates (gravel), building and dimension stones (limestones….)
- (Bio)-chemical precipitation
  - evaporite minerals (fertilizer)
  - Fe, Mn, P deposits
- Deposition and alteration
  - Organic matter, hydrocarbon source rocks
- Weathering processes
  - residual deposits (Ni, Al bauxite, clay deposits)

Energy and Resources
Summary weeks 2/3

- Fluids in the crust are essential to mobilize, transport and concentrate chemical components that constitute energy and resource commodities.
- Fluids are melts, aqueous and hydrocarbon fluids. They derive from melting of crust, exsolution from melts, compaction of clays, breakdown of hydrous minerals, and circulating surface waters.
- Fluid circulation is controlled by fluid storage (porosity), fluid flow capacity (permeability), thermal gradients and pressure. Tectonic forces play an eminent role, and via deformation of rocks can allow fluid movement.
- Commodity deposits can be classified according to fluids and key processes:
  - Magmatic
  - Magmatic-hydrothermal
  - Hydrothermal
  - Sedimentary and surface
- Note:
  - Individual commodities usually occur in more than one type of deposit
  - Single deposits can be polygenetic, with several overprinted processes